
E102 Inverted Pendulum Report

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I. Introduction

The inverted pendulum is an inherently unstable and nonlinear model. Without active control, the slightest movement will result in the pendulum falling over. In this experiment, a cart with an attached inverted pendulum is required to move **exactly** 1 meter away from its initial position in **10 or less seconds**. The cart is equipped with sensors that measure the angular displacement of the pendulum, $\theta(t)$, and the cart's position, $s(t)$. The pendulum is subject to an angular acceleration disturbance, $\alpha(t) = 0.5 \text{ rad/s}^2$, as well an acceleration limit, $|a(t)| < 0.5 \text{ m/s}^2$. This report will discuss the design process of the state feedback control system, the corresponding simulations, and how these were used to stabilize the system.

II. State Space Design of a Linear Controller for a Linearized Plant

To design the linear controller, the state space representation of the linearized system needed to be calculated. As shown in Figure 1, the small-angle approximation was utilized to linearize the describing equations, subsequently linearizing the plant.

$$\begin{array}{ccc}
 L \frac{d^2 \theta(t)}{dt^2} - g \sin \theta(t) = -a(t) \cos \theta(t) + L \alpha(t) & \xrightarrow{\text{Small Angle Approximation}} & L \frac{d^2 \theta(t)}{dt^2} - g \theta(t) = -a(t) + L \alpha(t) \\
 \frac{d^2 s(t)}{dt^2} = a(t) & & \frac{d^2 s(t)}{dt^2} = a(t)
 \end{array}$$

Figure 1: The system's describing equations before and after linearization. $L = 0.5 \text{ m}$.

As shown in Figure 2, the aforementioned equations can be used to determine the state-space equations. The system, input, and output matrices (A, B, and C) are particularly important, since they are used to solve for the observer gains and formulate the augmented matrices.

$$\begin{aligned}
 \dot{x} &= \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ g/L & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \\ s(t) \\ \dot{s}(t) \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 \\ -1/L \\ 0 \\ 1 \end{bmatrix}}_B a(t) + \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_E \alpha(t) \\
 y &= \underbrace{\begin{bmatrix} \theta(t) \\ s(t) \end{bmatrix}}_y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_C \underbrace{\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \\ s(t) \\ \dot{s}(t) \end{bmatrix}}_x
 \end{aligned}$$

Figure 2: The linearized state-space equations with matrices A, B, C, D, and E. $L = 0.5 \text{ m}$.

Finding the eigenvalues of A demonstrates that the system is unstable given that it has poles $s_1 = 0$ and $s_{2,3} = \pm 4.42719$. The rank of the system's controllability matrix,

$M_c = [B \ AB \ A^2 B \ A^3 B]$, and observability matrix, $M_o = [C \ CA \ CA^2 \ CA^3]^T$, are both non-zero, meaning that the system is both observable and controllable. The system can be stabilized using

a state feedback control system with integral action and an observer, as shown in the full system overview in Appendix B. Developing the integral action for the system requires constructing the augmented state space equations. As a result of the integral action, a new state variable representing the integral of the steady state error, x_i , is introduced to the state vector. The state vector for the augmented state equation becomes $\xi = [x_i, \theta(t), \theta'(t), s(t), s'(t)]^T$. Using Equation 1, where $A_a = [0, -C_{2,:}; 0, A]$, $B_a = [-D; B]$, $B_r = [1; 0]$, and $B_w = [-D; B]$, the augmented state equations can be formulated.

$$\xi' = A_a \xi + B_a u + B_r r + B_w w \quad (1)$$

Since the system cannot overshoot and must have a settling time of 10 seconds or less, the system must satisfy the following M_p and t_s equations: $e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} 0$ and $\frac{4.6}{\zeta\omega_n} \leq 10$. Solving these equations gives $\omega_n = 0.46$ rad/s and $\zeta = 0.99$ (slightly less than one to avoid dividing by zero).

Given ζ and ω_n , the system's desired poles are located at $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The poles of the new 5x5 augmented system matrix A_a are largely dominated by the desired poles, $s_{1,2}$, while the remaining three poles need to be negative, real, and decay faster than the calculated desired poles. The observer, which is based on the original state-space equations, will have four poles set by the designer that must be real, negative, and between 2-5 times greater than the desired poles. As shown in Appendix A, MATLAB's place command was used on A_a , B_a , and the augmented system's poles in order to solve for the feedback gain matrix K , while the place command was utilized on A^T , C^T , and the observer poles to solve for the observer gain vector L . Both the observer poles and system poles were arbitrarily tuned until the system's output met the requirements, as discussed in Section III.

III. Simulation of Control of Nonlinear Plant with the Linear Controller

The complete cart-pendulum control system was modeled and simulated in Simulink. Due to the nonlinear and inherently unstable nature of the cart-pendulum plant, the feedback controller with the augmented state equation designed in Section II was implemented along with a full-state observer to stabilize the system. Appendix B illustrates the overall control block diagram which includes the feedback controller, nonlinear cart-pendulum plant, and observer block in detail.

In order to implement the linear controller, estimates of the state variables derived from the measured outputs of the system and the system inputs needed to be made. This was done with an observer subsystem.

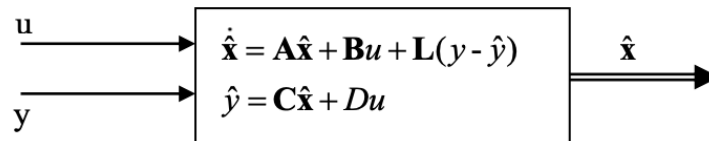


Figure 3: State equations governing observer design and implementation

The observer design was governed by the state equations shown in Figure 3, where A, B, C, and D were the system matrices derived from the linearization in Section II. $\mathbf{L}$ was the observer gain matrix determined through pole placement.

The control law applied to design the feedback controller was the following:

$$u = -[K_f \quad \mathbf{K}]\xi$$

Both the controller and observer gains were selected using pole placement. The final pole sets that yielded a stable system with no overshoot were:

$$\text{Controller poles} = [-0.4554 \pm 0.0649j, -5, -9, -13]$$

$$\text{Observer poles} = [-0.4554, -20, -36, -52]$$

Notice that the non-dominant observer poles are 4 times greater than the controller poles. The poles above correspond to the following gains respectively:

$$[K_f \quad \mathbf{K}] = [6.3092, -150.5141, -34.4389, -29.6051, -40.9671]$$

$$L = \begin{bmatrix} 59.0001 & -17.4702 \\ 688.486 & -506.266 \\ -16.7772 & 49.4553 \\ -478.3457 & 387.5519 \end{bmatrix}$$

These gains achieved fast convergence of the estimated states and smooth closed-loop performance. Control input saturation was enforced at $\pm 0.5 \text{ m/s}^2$ to reflect physical limits. Simulation results confirmed that the cart reached the 1-meter target within 10 seconds without overshoot. Figure 4 depicts the plots of cart position, pendulum angle, and acceleration. The plots demonstrate that all project specifications were met.

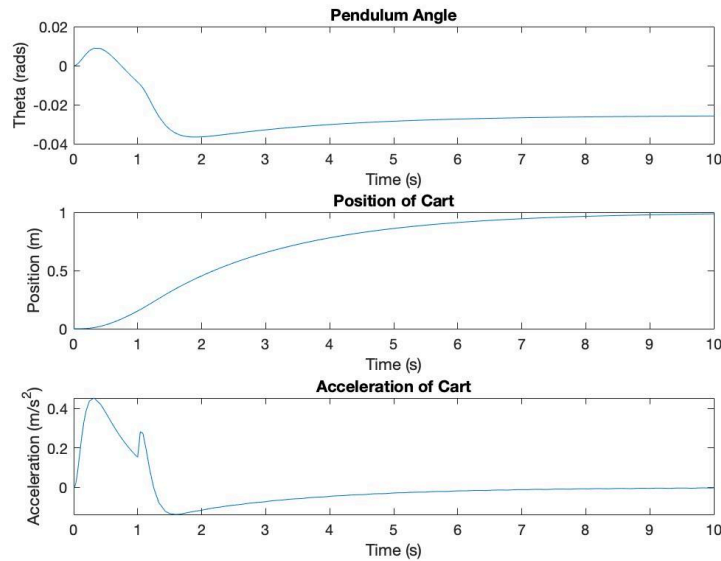


Figure 4: Plot of $\theta(t)$, $s(t)$, and $a(t)$ of the Cart-Pendulum Over A 10 Second Interval

IV. Discussion and Conclusions

A key aspect of this project was the careful tuning of closed-loop poles for both the state feedback controller and the full-state observer. The system's behavior—particularly in terms of settling time, overshoot, and stability—was highly sensitive to the placement of these poles.

The controller gains were tuned through several iterations of pole placement on the augmented system. Placing poles too close to the imaginary axis resulted in sluggish performance and increased pendulum deviation, while overly aggressive placement led to actuator saturation and degraded system stability. Through experimentation, a balanced configuration was identified that provided fast convergence without sacrificing robustness or violating physical constraints.

Observer poles were placed significantly farther left than the controller poles, in accordance with the common rule of thumb that observers should be 2 to 5 times faster than the feedback loop. This ensured rapid state estimation while avoiding amplification of sensor noise. Observers with poles too close to the controller led to lag in estimation, while excessively fast observers introduced instability.

Simulation results from Section III confirmed that the tuned system met all project requirements. The cart reached the 1-meter target smoothly, without overshoot. The pendulum angle remained close to vertical with minimal oscillations, and the control input stayed within the saturation bounds of $\pm 0.5 \text{ m/s}^2$. It was also observed that the system could not tolerate disturbances beyond the specified value of $\alpha(t) = 0.5 \text{ rad/s}^2$. Higher disturbance values resulted in loss of stability.

These outcomes demonstrate the effectiveness of state-space design with integral control and observer feedback, and highlight the importance of thoughtful pole tuning in achieving a stable, high-performance control system.

Appendix A: Matlab Code

```
% System Parameters
g = 9.81;
l = 0.5;
zeta = 0.99;
wn = 0.46;
alpha = 0.5;

% Define State Space Matrices
A = [0 1 0 0;
     g/l 0 0 0;
     0 0 0 1;
     0 0 0 0];
B = [0;-1/l;0;1];
C = [1 0 0 0;
     0 0 1 0];

% Augmented State Matrices
Aa = [0 0 0 -1 0;
      0 0 1 0 0;
      0 g/l 0 0 0;
      0 0 0 0 1;
      0 0 0 0 0];
Ba = [0;0;-1/l;0;1];

% Closed Loop Poles
pk = [-zeta*wn+j*wn*sqrt(1-zeta^2) -zeta*wn-j*wn*sqrt(1-zeta^2)
      -5 -9 -13];
pl = [-zeta*wn -4*5 -4*9 -4*13];
Kgain = place(Aa, Ba, pk);
Ki = Kgain(1);
K = Kgain(2:5);
L = place(A',C',pl)';

% Simulink
simout =
sim('InvertedPendulumSimulation.slx','ReturnWorkspaceOutputs','o
n');

% Plots of Theta, Position, and Acceleration
t = simout.tout;
theta = simout.yout(:,1);
s = simout.yout(:,2);
a = simout.yout(:,3);

figure(1)
```

```

plot(t,theta);
title('Pendulum Angle')
xlabel('Time (s)')
ylabel('Theta (rads)')

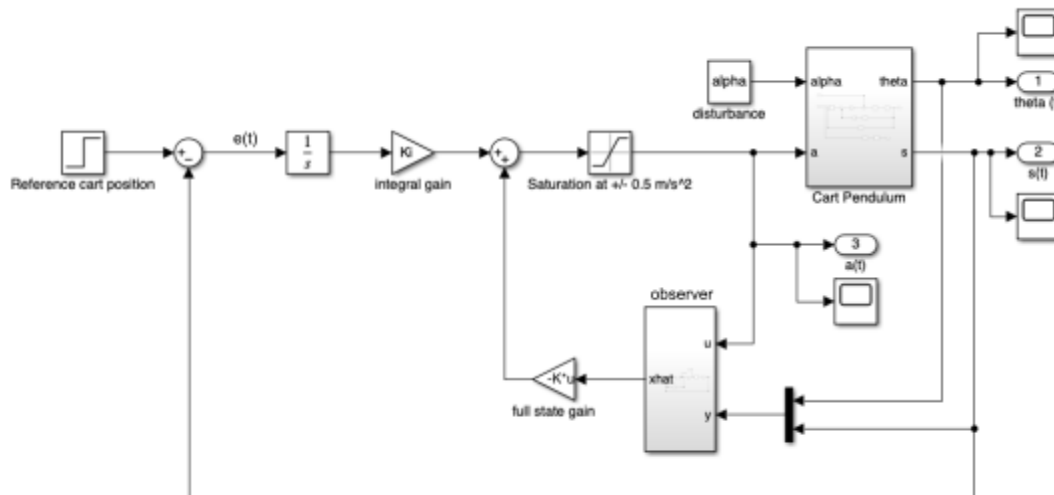
figure(2)
plot(t,s);
title('Position of Cart')
xlabel('Time (s)')
ylabel('Position (m)')

figure(3)
plot(t,a);
title('Acceleration of Cart')
xlabel('Time (s)')
ylabel('Acceleration (m/s^2)')

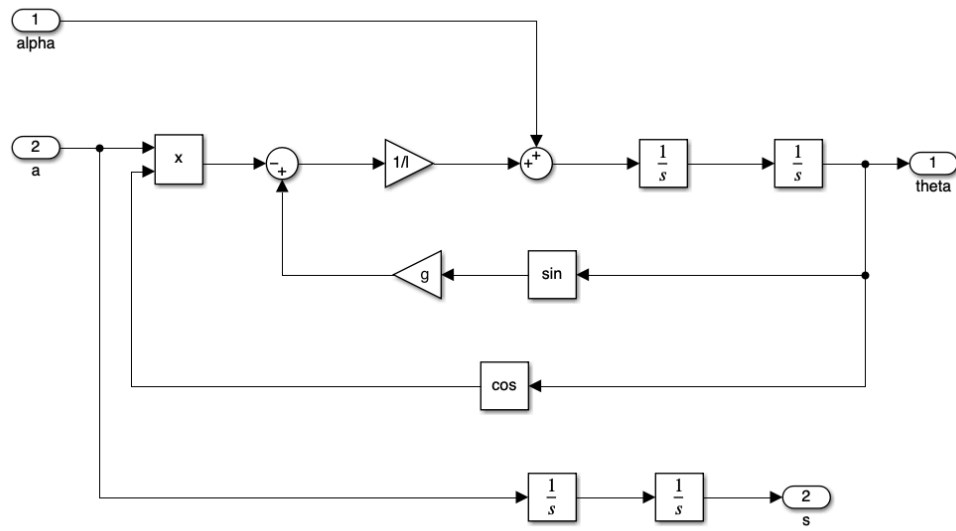
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Appendix B: Simulink Block Diagram

Overall System Block Diagram



Cart Pendulum



Observer

